Abbavaram Gowtham Reddy

Postdoctoral Researcher at CISPA Helmholtz Center for Information Security, Saarbrücken, Germany

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Research Summary

I develop causality-grounded machine learning models for generalization, interpretability, and trustworthy decision-making. My research interests are: generalization under hidden confounding shift, causal machine learning for interpretability, and causal reasoning in foundation models.

- Explaining why boosting succeeds and invariance learning fails under hidden confounding [**2 × ICLR 2026**]
- Causal modeling of neural networks and developing inherently explainable (ante-hoc) neural network models [**AAAI 2024, ICML 2022**]
- Developed tools for causal inference using large language models and for improving causal inference capabilities of large language models from axiomatic training [**ICLR 2025, ICML 2025**]
- Developed diagnostic tools for causal representation learning and statistical causal inference in the presence of hidden confounders [**NeurIPS 2024, AAAI 2024, AAAI 2022**]

Current Affiliation

Postdoctoral Researcher CISPA Helmholtz Center for Information Security, Saarbrücken, Germany, Advisors: Dr. Krikamol Muandet and Dr. Rebekka Burkholz — Oct 2024 – Present

Publications

2026

- Amine M'Charrak, Abbavaram Gowtham Reddy, Thomas Lukasiewicz, Michael M. Bronstein, Krikamol Muandet, Fiedler-Regularized Causal Discovery for Sparse Connected DAGs, Conference on Neural Information Processing Systems (**NeurIPS**) 2026.
- Kiet Q. H. Vo, Abbavaram Gowtham Reddy, Julian Rodemann, Siu Lun Chau, Krikamol Muandet, Off-Policy Evaluation with Strategic Agents via Local Disclosure, International Conference on Machine Learning (**ICML**) 2026.
- Abbavaram Gowtham Reddy, Celia Rubio-Madrigal, Rebekka Burkholz, Krikamol Muandet, When Shift Happens - Confounding Is to Blame, International Conference on Learning Representations (**ICLR**) 2026.
- Abbavaram Gowtham Reddy, Rajeev Verma, Celia Rubio-Madrigal, Krikamol Muandet, Rebekka Burkholz, Boosting for Predictive Sufficiency, International Conference on Learning Representations (**ICLR**) 2026.

2025

- Hari Chandana Kuchibhotla, Sai Srinivas Kancheti, Abbavaram Gowtham Reddy, Vineeth N Balasubramanian, Efficient Vocabulary-Free Fine-Grained Visual Recognition in the Age of Multimodal LLMs, Transactions on Machine Learning Research (**TMLR**) 2025.
- Aniket Vashishtha, Abhinav Kumar, Atharva Pandey, Abbavaram Gowtham Reddy, Kabir Ahuja, Vineeth N. Balasubramanian, Amit Sharma, Teaching Transformers Causal Reasoning through Axiomatic Training. International Conference on Machine Learning (**ICML**) 2025.
- Aniket Vashishtha, Abbavaram Gowtham Reddy, Abhinav Kumar, Saketh Bachu, Vineeth N. Balasubramanian, Amit Sharma, Causal Order: The Key to Leveraging Imperfect Experts in Causal Inference. International Conference on Learning Representations (**ICLR**) 2025.
- Hari Chandana Kuchibhotla, Sai Srinivas Kancheti, Abbavaram Gowtham Reddy, Vineeth N Balasubramanian, Semantic Alignment for Prompt-Tuning in Vision Language Models, Transactions on Machine Learning Research (**TMLR**) 2025.

2024

- Abbavaram Gowtham Reddy, Vineeth N Balasubramanian, Detecting and Measuring Confounding Using Causal Mechanism Shifts. Conference on Neural Information Processing Systems (**NeurIPS**) 2024.
- Abbavaram Gowtham Reddy, Vineeth N Balasubramanian, NESTER: An Adaptive Neurosymbolic Method

for Causal Effect Estimation, AAAI Conference on Artificial Intelligence (**AAAI**) 2024.

- Abbavaram Gowtham Reddy, Saketh Bachu, Harsharaj Pathak, Benin Godfrey L, Varshaneya V, Sathyanarayan Kar, Vineeth N Balasubramanian, Towards Learning and Explaining Indirect Causal Effects in Neural Networks. AAAI Conference on Artificial Intelligence (**AAAI**) 2024.

2022

- Abbavaram Gowtham Reddy*, Sai Srinivas Kancheti*, Vineeth N Balasubramanian, Amit Sharma, Matching Learned Causal Effects of Neural Networks with Domain Priors. International Conference on Machine Learning (**ICML**) 2022.
- Abbavaram Gowtham Reddy, Benin Godfrey L, Vineeth N Balasubramanian, On Causally Disentangled Representations, AAAI Conference on Artificial Intelligence (**AAAI**) 2022.
- Abbavaram Gowtham Reddy*, Saloni Dash*, Amit Sharma, Vineeth N Balasubramanian, Counterfactual Generation Under Confounding, NeurIPS workshop on CML4Impact, 2022.

2021

- Abbavaram Gowtham Reddy, Benin Godfrey L, Vineeth N Balasubramanian, CANDLE: An Image Dataset for Causal Analysis in Disentangled Representations, **Best paper award winner** at CVPR 2021 workshop on Causality in Vision.
- Abbavaram Gowtham Reddy, Causality in Neural Networks, Extended Abstract AIES 2021.

Education

Ph.D.	Indian Institute of Technology Hyderabad, India • Advisor: Prof. Vineeth N Balasubramanian • Thesis: Bridging Causal Inference and Neural Network Models: Methods and Applications • Coursework CGPA: 9.38/10.0	Aug 2019 – Sep 2024
M.Tech	JNTU Hyderabad, India • Computer Science and Engineering, CGPA: 7.78/10.0	Sep 2013 – Sep 2015
B.Tech	JNTU Anantapur, India • Computer Science and Engineering, CGPA: 7.37/10.0	Apr 2008 – Apr 2012

Awards, Honors, and Fellowships

- Top Reviewer, Conference on Neural Information Processing Systems (NeurIPS) 2026
- Silver Reviewer, International Conference on Machine Learning (ICML) 2026
- Top Reviewer, Conference on Neural Information Processing Systems (NeurIPS) 2025
- **$$** DAAD AI-Net Fellowship for Postdoctoral Networking in Artificial Intelligence, Germany (2024)
- **$$** Prime Minister's Research Fellowship (**PMRF**) India, Sep 2020 - Jul 2023 (PMRF ID: 2000550)
- Research Excellence Award, IIT Hyderabad (academic year 2024 - 2025)
- Research Excellence Award, IIT Hyderabad (academic year 2022 - 2023)
- Fearless Problem Solver, EPAM Systems India, 2018
- UGC-NET Junior Research Fellowship, 2016, 2015
- UGC-NET Assistant Professorship, 2015, 2014, 2013

Grants and Funding

- **$$$** Contributed to a successful Microsoft Research grant proposal, resulting in **$20,000** in AI compute credits.

Research Services

- Co-organizer of the Rational Intelligence Seminar Series (Jun 2025 - Present) at CISPA
- Reviewer: NeurIPS (Top reviewer 2026 & 2025), ICML (Silver Reviewer 2026), ICLR, AAAI, UAI, AISTATS, CVPR, WACV, ACML, TMLR
- Student volunteer for Asian Conference on Machine Learning (ACML) 2022

Invited Talks

"Predictive Sufficiency for Generalization Under Hidden Confounding Shift" — presented as part of the seminar series "AI-Powered Innovation: Integrating Future-Ready Technologies in Real-World Applications". NIT Jalandhar, India, Nov 2025

Research Forums Attended and Presentations

- Poster presentation at International Conference on Machine Learning, July 2026, Seoul, South Korea
- Poster presentation at International Conference on Learning Representations - Apr 2026, Rio de Janeiro, Brazil
- Dagstuhl Seminar on Causality and Large Language Models: Opportunities to Advance Causal Reasoning - April 2026, Wadern, Germany
- Conference on Neural Information Processing Systems - Dec 2025, San Diego, USA
- Poster presentation at Google Research Week - Feb 2024, Bengaluru, India
- Poster presentation at Amazon Research Days - Dec 2023, Bengaluru, India
- Google Research Week - Jan 2023, Bengaluru, India
- Poster presentation at Amazon Research Days - Nov 2022, Bengaluru, India

Teaching and Mentoring

- Co-advising a bachelor's thesis on imprecise probabilistic causal inference
- Teaching assistant, University of Saarland: Advanced Topics in Rational Intelligence (2026).
- Teaching assistant, IIT Hyderabad: Deep Learning for Computer Vision (2020), Causal Inference (2021), Introduction to Programming (2023)
- Taught the following courses as part of the PMRF fellowship: Mathematical Foundations of Machine Learning at BVRIT, Hyderabad (2022), Machine Learning at PSG College, Coimbatore (2021), and Introduction to Computer Science at SV College of Engineering, Tirupati (2021)
- Delivered several tech talks on machine learning and Python, EPAM system India (2015–2018)

Professional Experience

- Software Engineer at EPAM Systems India, Hyderabad. Sep 2015 - Jul 2018
- Research Intern at Microsoft Research India, Bengaluru. Jul 2022 - Oct 2022